



Towards Stereoscopic On-vehicle AR-HUD

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Abstract

On-vehicle AR-HUD (head-up display) is a driving assistant system, which can let drivers see navigation and warning information directly through the windshield with an easy-to-perceive augmented reality form. Traditional AR-HUD can only produce the content at a specific distance, which causes uncomfortable experiences such as frequent refocusing and unnatural floating of virtual objects. This paper proposed an innovative AR-HUD system, which can provide stereoscopic scenes to the driver. The system is composed of two traditional HUD displays and supports parallax by additive light field factorization. Optical paths and illumination of two displays are precisely calibrated for both views to ensure the combination of lights is as expected. The factorization algorithm is optimized for the system. With GPU acceleration, the system can run in real time. The system is cheap and simple to transform from a traditional AR-HUD system, which presents a feasible scheme to achieve a fusion-enhanced augmented reality assistant for driving.

Keywords Stereoscopic · Light field · Augmented reality · Head-up display · Calibration

1 Overview

Head-up display (HUD) is a transparent display that can present information in front of the user's eyes. The origin of the name "Head-up" stems from a pilot being able to view information with the head positioned "up" and looking forward, instead of angled down looking at lower instruments. Although initially developed for military aviation, HUD has penetrated various fields which have requirements of keeping eyes focused, like commercial automobiles. AR-HUD uses augmented reality techniques to present information in an immersive form, like laying virtual guide paths on the road, highlighting noteworthy pedestrians, and sticking interesting labels on roadside buildings. AR-HUD further improves intuitiveness, ease of use, and security.

However, state-of-the-art AR-HUD, although has contents registered and rendered in 3D space, can only present them in 2D images like a transparent screen put in front of the viewer because of lacking parallax. An example is that during the experience of an on-vehicle AR-HUD prototype, many participants complained about the arrows of guide path pointing

to the sky, which were supposed to be laid on the ground. This effect can be eliminated by watching the image with only one of their eyes. This means despite that AR-HUD presents images at a relatively large distance from the viewer, stereoscopic still plays an important role in experience of augmented-reality fusion.

Inspired by previous works on glasses-free stereoscopic displays [15], we proposed a novel stereoscopic on-vehicle AR-HUD prototype that enables parallax without auxiliary equipment like glasses. As shown in Fig. 1, the prototype consists of two HUDs with different imaging distances, whose images are optically combined when seen by the viewer. To generate the desired binocular light field, an additive light field factorization method is implemented, which determines what contents should be shown on each HUD screen. This schema and process are similar to [15], but the complex optical path in HUD brings serious and irregular distortion. Moreover, the deformation of distortion related to the view-point is not negligible when the eyebox expands to binocular size. (The borders of gray quads and white corner markers in Fig. 1d and e reflect the distortions.) To solve these problems, we proposed an optical path calibration method and an optimized light field factorization algorithm to solve the problem. Furthermore, to make the system practically useful, the distortion correction and light field factorization are

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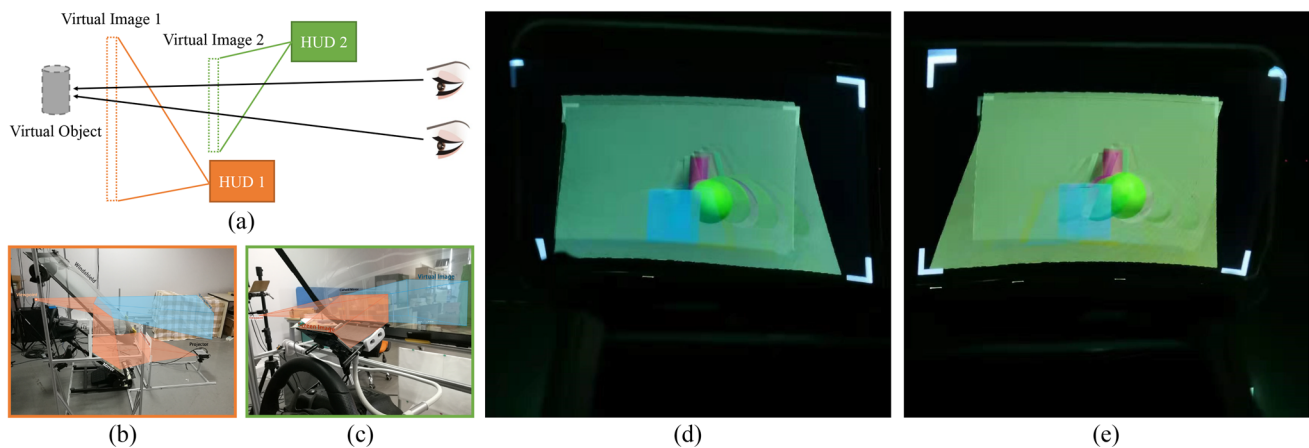


Fig. 1 Our stereoscopic AR-HUD prototype (a) contains two traditional HUDs (b) and (c) with different imaging distances. d and (f) are the binocular images captured by a camera put at the positions of viewer's left and right eye, respectively

accelerated using GPU to meet the requirement of real-time performance.

Our main contributions in this paper include:

- A novel AR-HUD scheme that can produce glasses-free stereoscopic images.
- A real-time additive light field factorization applying for proposed AR-HUD scheme.
- An effective calibration and correction methods for image distortion in AR-HUD.

2 Related Work

2.1 Head-Up Display

Lots of researches focused on the application of AR-HUD for vehicles. Park et al. [23] developed a real-time system that visualizes information to tower crane drivers to increase the safety of the construction field. Park et al. [21] investigated the efficient method to provide traffic information to drivers using an AR-HUD system. They also proposed an AR HUD system to enhance driving safety under bad weather conditions [22]. Yoon et al. [32] presented a HUD vehicle navigation system, which acquires lane information from a camera on top of the vehicle and the upcoming turning information from a pre-equipped navigation map. Merenda et al. [18] investigated which color can competent with on-vehicle AR-HUD. Chiu et al. [2] proposed an approach that utilizes semantic information to register 2D monocular video frames to the world using 3D georeferenced data for AR driving applications. Lagoo et al. [13] proposed a system combined with HUD and gesture recognition. But this system is not well integrated with AR. This kind of works uses relatively simple hardware structure for their HUD prototypes, such as a see-

through transparent liquid-crystal display (LCD) installed in the windshield to show information [18,21], which provides uncomfortable focusing distance and shows 2D images only. Our work has an objective orthogonal to these works, which aims at designing and implementing a stereoscopic HUD. It can provide the viewer 3D experience through stereopsis and more comfortable viewing.

2.2 Light Field Display

Our idea of enabling stereoscopic capability is inspired by light field displays. The concept "light field" was first proposed by Gershun [6,7] in 1936. Light field display provides a more accurate converge for 3D viewing and more accurate depth information [14]. Yoshida et al. [33] were first introduced light field to AR/VR displays. They proposed a tabletop AR display using light-field reproduction. But the images displayed are somewhat unclear and unfocused. Wetzstein et al. [28] implemented a light field display with stacked LCD planes to support multi-view capability. Huang et al. [9] proposed a computational display technology for people who need glasses to see properly using the light field. This technology pre-distorts the present contents and achieves significantly high resolution and contrast than prior vision-correcting image display. Their following work [8] designed a pair of VR glasses which presents a factored light field for each eye. This prototype supports high image resolution and provides more natural viewing experiences than conventional near-eye displays. Later, Lee et al. [15] proposed a see-through additive light field display using holographic optical elements, achieving bright and full-color volumetric images in high definition. Zhan et al. [34] proposed a 3D near-eye display system that presents different images with real depths for each eye. Therefore, this system provides approximate focus cue and relieve the dis-

comfort from vergence–accommodation conflict. Despite the great work done by these researches, implementing a stereoscopic HUD system through light field technology still meets challenges such as non-planar display layers and complex optical distortion. Our work made a first step towards these challenges through a precise calibration and an improved factorization.

2.3 Calibration

Calibration of AR-HUD is closely related to calibration methods proposed for optical see-through head-mounted display (OST-HMD) because OST-HMD and AR-HUD have similar optical structure. In this section, we discuss works about calibrating OST-HMD first, followed by calibrating AR-HUD.

The OST-HMD calibration generally contains two parts. First, the projection properties of OST-HMD are estimated and represented as a projection matrix. Many classical calibration methods such as SPAAM [26] and stereo SPAAM [5] are based on user's interaction to align the virtual image points with 3D markers in the real world from various viewpoints with the help of magnetic or optical tracking systems. These methods are practical in industrial cases and can be seen in off-the-shelf AR products. Then Yuta Itoh et al. [10] and Plopski et al. [24] came up with INDICA and CIC with the idea of eye-tracking to do continuous recalibration with the help of an extra camera. With the miniaturization of depth cameras, the integration of OST-HMD and depth camera can be achieved. The external tracking system can be replaced by simpler systems. Jun and Kim [11] proposed such a calibration system with less user's repetitions. A hand and gesture tracking device such as leap motion can also be helpful to the calibration of OST-HMD [19]. Secondly, distortion of the projector system, which is defined as the difference between the original image being displayed and the final image perceived by the user, must be corrected. There are two kinds of computational correction methods: camera calibration methods and image correction methods. Camera calibration methods directly estimate the parameters of HMD. Owen et al. [20] obtained parameters by running Tsai [25] camera calibration algorithm twice. The other kind of methods disregard the complex optical system and directly consider the relationship between the standard image and distorted image. The distortion can be modeled as a spatial mapping function defined in a two-dimensional image coordinate using global methods [1] or local methods [16].

Despite the similarity, there are significant differences between AR-HUD and OST-HMD. AR-HUD has a much larger eyebox (the region of feasible viewing position) than OST-HMD and the irregular reflective surfaces involved in AR-HUD causes a more complex distortion pattern. Thus calibrating AR-HUD is a more challenge task than calibrating

OST-HMD. Wu et al. [30] built an indoor system prototype to simulate the actual driving situation on the road. They only considered the distortion correction at a fixed viewing position. They provided a primitive idea of augmenting the reality by 2D overlaying but lacking 3D registration of virtual world in the real world, which makes it unpractical for real car AR applications. Wientapper et al. [29] decomposed the calibration into two phases: view-independent geometry of the virtual plane and view-dependent image warping which is similar to the display-relative calibration method proposed by Owen et al [20]. For correcting the distortion, they employed a higher-order polynomial function of 5 parameters. They used a vision-based tracking method by attaching textured patterns on the windshield to achieve accurate camera registration with the help of structure-from-motion techniques. The setup was cheap but time-consuming during the preparation. Instead of using a polynomial function, Li et al. [17] achieved comparable precision with the help of a multilayer feed-forward neural network model. Ueno and Komuro [27] took the diversity of road surface and practical face tracking into consideration. They used a depth camera to measure the road surface in real time and a camera to perform face tracking to reduce the effect of head movement. Xia et al. [31] proposed a 3D calibration method for auto-stereoscopic multi-view displays. They used gray code patterns to modulate the images shown on the display, accelerating the process of calculating the correspondence relationship between the display and the locations of the capturing camera. Gao et al. [4] used a consumer-grade mono-camera to achieve the calibration and also took eye tracker into account. Because the align process they use is time-consuming, only 9 viewpoints are calibrated and other viewpoints are done by linear interpolation.

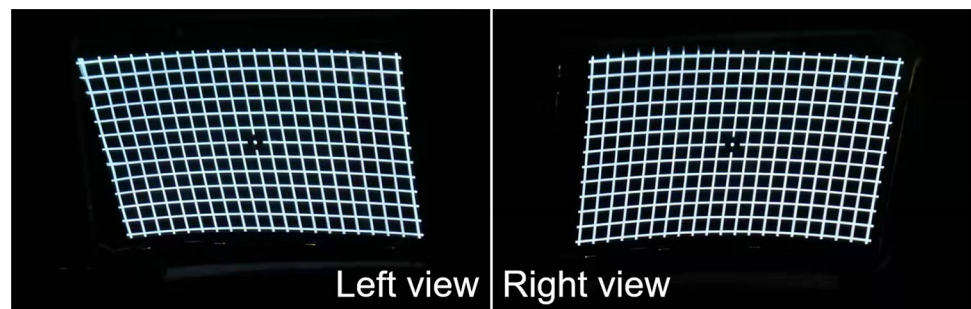
3 Hardware Scheme

3.1 General Structure of HUD

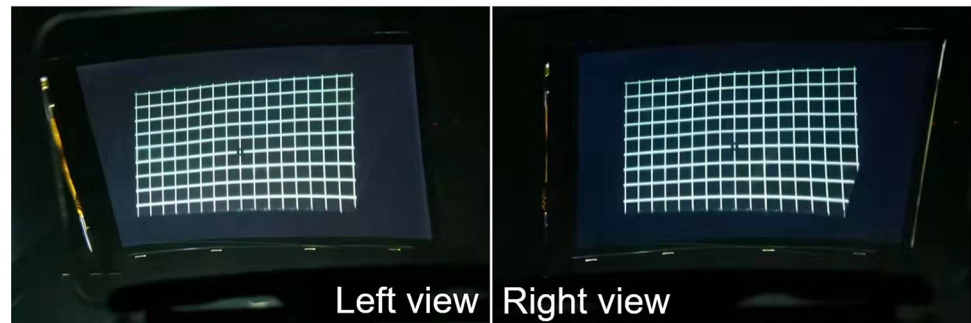
A general HUD consists of a LCD display (or micro projector) and one or more mirrors to reflect the displayed image into viewer's eye. Concave reflective surfaces are involved to enlarge the image and push it away to a proper distance. The downside is that such surfaces introduce image distortion. The distortion is significant and irregular especially when the system contains several concave surfaces or freeform surfaces like the windshield (Fig. 2). To correct the distortion of image and further register the image with environment, a calibration process needs to be performed, which is described in Sects. 5 and 6.

Such a display system can produce image at a specific distance. But without parallax, viewer will have strong feeling of virtual elements misaligned with environment, especially

Fig. 2 The distortion of virtual images shown by HUDs. **a** Factory-installed HUD; **b** portable HUD. Photographs taken from left eye position and right eye position show the differences of distortion at these two viewpoints



(a)



(b)

for those elements laying on the ground. Another drawback is that driver need to refocus frequently between environment and HUD's image, which causes eye fatigue and security risks. To deal with the problem of lacking parallax, we proposed a stereoscopic HUD schema.

3.2 Stereoscopic HUD

Our stereoscopic HUD consists of two sets of HUD hardware. The virtual images of two HUD components overlap with each other but have different distance (Fig. 1a). When seen by viewer, two images are superimposed together. In this paper, we assume that viewer's eyes are fixed, but our system can be simply extended to support movement of eyes with an commercial eye tracker. To enable stereopsis, the HUD should produce different images for each of the viewer's eyes. This can be achieved because at different viewpoint, different pixels of two displays are combined as shown in Fig. 1a. The contents shown on two displays can be solved using additive light field factorization described in Sect. 4. In this paper, we built a prototype using a factory-installed-like HUD (Fig. 1b) and a portable HUD (Fig. 1c). The factory-installed-like HUD has several reflective surfaces and takes vehicle's windshield as final reflective component, displaying its image at about 5 m forward. The portable HUD uses only one curved transparent glass, so its imaging distance is relatively small, which is about 1.5 m. So our prototype

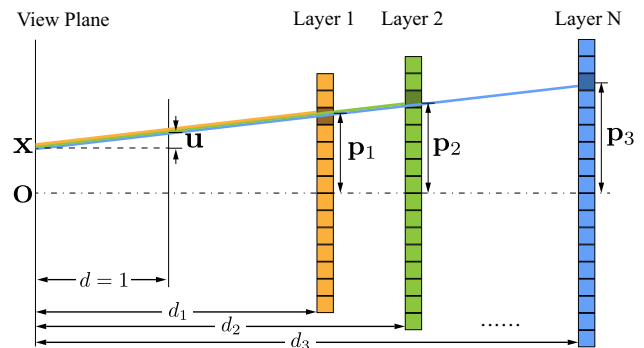


Fig. 3 Typical schematic of multi-planar-layers display

can provide proper generation of stereoscopic images for elements with depths between 1.5 and 5 m.

4 Additive Light Field Factorization

4.1 Multi-Planar-Layers Display

We first discuss the general additive light field factorization method for the multi-planar-layers display. Figure 3 shows a typical schematic of a multi-planar-layers display. A light field can be described as a 4D function $L(x, u)$, where $x \in \mathbb{R}^2$ is the intersection of a light ray on view plane, and $u \in \mathbb{R}^2$

is the angular measurement at the plane with distance of 1 unit. The ray passes through N transparent display planes at distances $\{d_n\}$. The intersection of ray $(\mathbf{x}, \mathbf{u})$ and the n^{th} display plane is:

$$\mathbf{p}_n = \mathbf{x} + d_n \mathbf{u} \quad (1)$$

For an additive light field display, light emitted from the display planes is combined before entering the views' eye (e.g., through an optical combiner):

$$L(\mathbf{x}, \mathbf{u}) = \sum_{n=1}^N I_n(\mathbf{p}_n) = \sum_{n=1}^N I_n(\mathbf{x} + d_n \mathbf{u}) \quad (2)$$

where I_n is the image shown on the n^{th} display plane.

The light field factorization problem can be described as follows: solving the images $\{I_n\}$ that best reconstruct a given light field L_0 . This problem can be formulated as an optimization equation:

$$\begin{aligned} \min \quad & \|L_0(\mathbf{x}, \mathbf{u}) - \sum_{n=1}^N I_n(\mathbf{x} + d_n \mathbf{u})\|^2 \\ \text{s.t.} \quad & 0 \leq I_n \leq 1, \forall n \in [1, N] \end{aligned} \quad (3)$$

Equation (3) is usually discretized to matrix-product form for computers to solve [15,34]. Assume that the discretized representation of target light field L_0 has angular resolution of V (i.e., V discretized values of $\mathbf{x}$) and its spatial resolution is D . The resolution of the n^{th} display plane is D_n . The discretized form of Eq. (3) can be turned into a bounded linear least squares problem:

$$\min \|\mathbf{L}_0 - \mathbf{P}\mathbf{I}\|^2 \quad \text{s.t.} \quad \mathbf{I} \in [0, 1] \quad (4)$$

where $VD \times 1$ vector $\mathbf{L}_0$ is vectorization of discretized target light field L_0 , $\sum_n D_n \times 1$ vector $\mathbf{I}$ is vectorization of all display planes. $VD \times \sum_n D_n$ matrix $\mathbf{P}$ is the mapping matrix represents the contribution of pixels on every display planes to a discretized light ray in the light field, which can be calculated using Eq. (1). Eq. (4) can be solved by many existed methods, such as [3].

4.2 Factorization for Stereoscopic HUD

Our stereoscopic HUD is composed of two display layers at different distance. because of the non-planar reflective elements existing in the optical system, the virtual images seen by viewer are non-planar surfaces, containing high-order distortion (Fig. 2). So Eq. (1) becomes a nonlinear function

$$\mathbf{p}_n = \chi_n(\mathbf{x}, \mathbf{u}) \quad (5)$$

The relationship between images $\{I_n\}$ and the displayed light field L (Eq. 2) turns into:

$$L = \sum_{n=1}^N I_n(\chi_n(\mathbf{x}, \mathbf{u})) \quad (6)$$

We also define the inverse function of χ_n as:

$$\mathbf{u} = \psi_n(\mathbf{x}, \mathbf{p}_n) \quad (7)$$

$\{\chi_n\}$ and $\{\psi_n\}$ depends on the optical structure of the HUD, so generally they cannot be expressed in analytic form. So instead of computing the matrix $\mathbf{P}$ and solve Eq. (4), we solve the problem taking advantage of GPU render pipeline(Fig. 5). The contents of displays are updated alternately using following rule:

$$I_n(\mathbf{p}) \leftarrow \text{Clamp} \left(I_n(\mathbf{p}) + \alpha \sum_v (L_0 - L)(\psi_n(\mathbf{x}_v, \mathbf{p})) \right) \quad (8)$$

where α is the update rate, which is set to 0.3 in our experiments. The function $\text{Clamp}(\cdot)$ truncates the calculated values to $[0, 1]$. Note that we still have difficulty in calculating L by Eq. (6) directly because $\{\chi_n\}$ is hard to evaluate. We in turn use the idea of device simulation and generate L through rendering. The implementation detail is described in Sect. 6.

5 Modeling HUD

As a HUD probably contains multiple non-planar reflective surfaces, it is hard to model the optical system in an analytical way. We treat the HUD as a black-box and don't care about its structure, thus keeping the calibration process general. Also note that we don't split the model into extrinsic parameters, projection parameters and distortion parameters, which are commonly used in HMD calibration. Specifically, the model builds an end-to-end mapping between the screen pixels and the light rays entering the viewer's eye, as shown in Fig. 4. The mapping is further related to the viewing position, so a relationship between the viewing position and mapping should be built.

5.1 Static Mapping

We first consider a static viewing position $\mathbf{x}$ in eye box. To correctly overlay virtual content on real scene, the directions of light rays emitted by pixels on HUD's screen should be measured at the viewpoint. As shown in Fig. 4, a predefined grid pattern (whose cross points are $\mathbf{p}_i$) is displayed on the screen and captured by a camera placed at $\mathbf{x}$. The cross points in the captured image are detected automatically. Let $\mathbf{P}_{\text{cam}}$

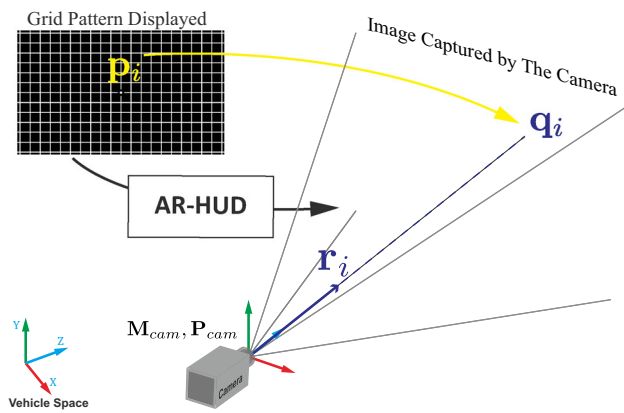


Fig. 4 Relationship between light ray $r(p_i)$, cross point in camera image q_i and cross point on HUD's screen p_i

be the camera's projection matrix and M_{cam} be the camera's transform matrix, we can calculate the corresponding light ray r_i of the i^{th} cross point from the detected coordinate q_i in the captured image:

$$r_i = M_{cam} P_{cam}^{-1} q_i \quad (9)$$

The dimensional of r can be further reduced by normalizing along Z axis:

$$u = \left(\frac{r}{r.z} \right) .xy \quad (10)$$

For a p between cross points, its corresponding light ray u is solved by linearly interpolating the four cross points around it. So for a static viewing position x , the end-to-end mapping is:

$$u = \psi^x(p) \quad (11)$$

Given the viewing position x and its pixel-light mapping ψ^x , the image displayed on the HUD can be corrected by an image wrapping process:

$$\text{Image}_{out}(p) = \text{Image}_{in}(P_{render} \psi^x(p)) \quad (12)$$

Where Image_{in} is the image rendered by a virtual render camera placed at x and P_{render} is the projection matrix of the render camera.

5.2 Handling Dynamic Viewing Position

As Eq. (7) defined, ψ is related to not only the position of pixel p but also the viewing position x . The most direct way is to uniformly take samples in the eyebox and do interpolation at runtime, similar to what [12] did for optical see-through glasses. But the eyebox of a HUD is much larger than that of

a near-eye display, which makes sampling a time-consuming task and results a trade-off between precision and quantity of work (determined by grid density). Also it is hard to extrapolate a reasonable result when the eye moves a bit out of the sampled range.

In this paper, we use polynomial function to model the relationship, inspired by [29]. Considering a pixel p on the screen, the mapping $\psi_p(x)$ is modeled using a polynomial function:

$$\psi_p = \sum_{d_1, d_2, d_3} b_p(d_1, d_2, d_3) x_x^{d_1} y_x^{d_2} z_x^{d_3}, \quad d_j \in \mathbb{N}, \sum_j d_j \leq D \quad (13)$$

where $b_p(\cdot)$ are coefficients of the polynomial function, D is the maximal polynomial degree of the function, which is device-related and determined by experiments.

To train the model, several hundreds of samples at different viewing positions in the eyebox are taken. After that, $b_p(\cdot)$ are calculated by solving the following optimization problem:

$$b_p(\cdot) = \underset{b_p(\cdot)}{\operatorname{argmin}} \sum_x \|u^x - \psi_p(x)\|^2 \quad (14)$$

Equation (14) is a standard least square problem, which can be solved by various methods. The complete mapping function $\psi(x, p)$ required by Eq. (8) can be evaluated by first evaluating all $\{\psi_{p_i}(x)\}$ at cross points of the grid pattern using Eq. (13) then interpolating p linearly between them.

6 Implementation

6.1 Hardware and Software Configuration

All routines are run on a desktop whose CPU is Intel Xeon E3-1231 and GPU is NVIDIA GeForce GTX 980. Source code is written using C# and Unity shader. The target light field is rendered using Unity with resolution of $V = 2$ and $D = 640 \times 480$. A prototype was built using a factory-installed like HUD (HUD1) and a portable HUD (HUD2) as shown in Fig. 1. The resolution of HUD1 is 1280×720 and the resolution of HUD2 is 800×480 . A simulation environment is built in Unity scene using the same configuration as our prototype.

6.2 Render-Based Factorization Pipeline

This section describes the pipeline of render-based additive light field factorization for the stereoscopic HUD. The pipeline consists of four steps, as shown in Fig. 5. The first

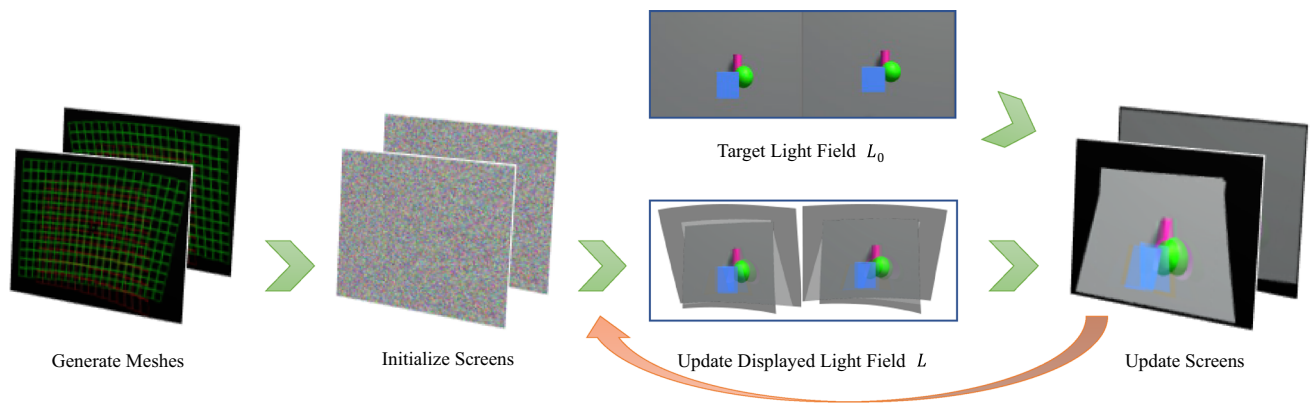
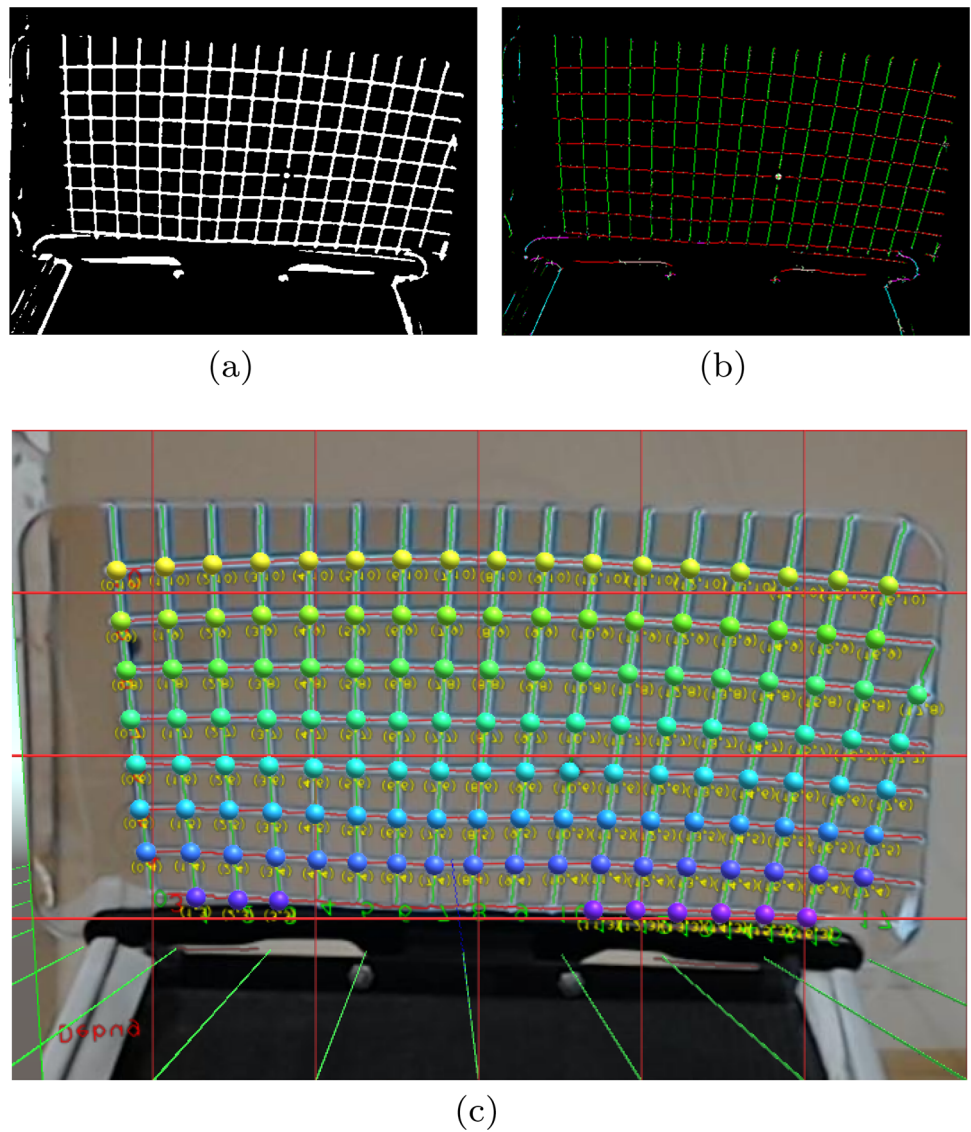


Fig. 5 The flow of light field factorization for binocular stereo HUD

Fig. 6 The steps of grid pattern detection method. **a** Adaptive threshold process extracts strong borders. **b** Line segments are detected and classified into horizontal type and vertical type. **c** Detected line segments are sorted and their cross points are calculated



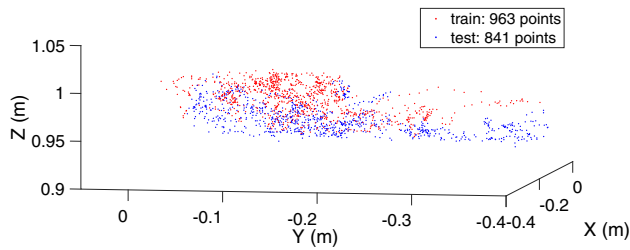


Fig. 7 The viewing positions of samples in train dataset and test dataset

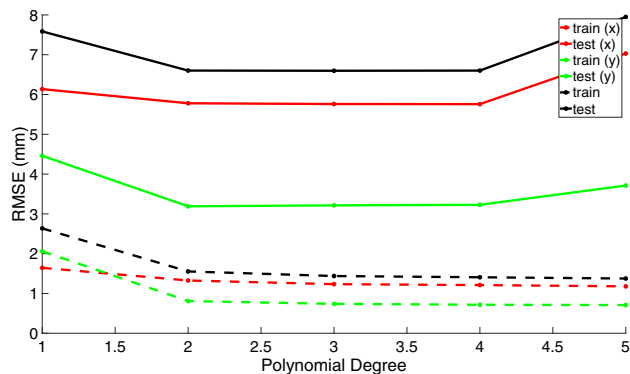
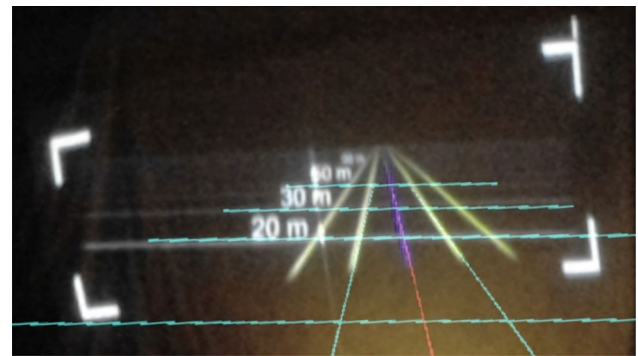


Fig. 8 The average error of models trained with polynomial degrees taken from 1 to 5. Dashed lines are the errors evaluated using train data, and solid lines are the errors evaluated using test data

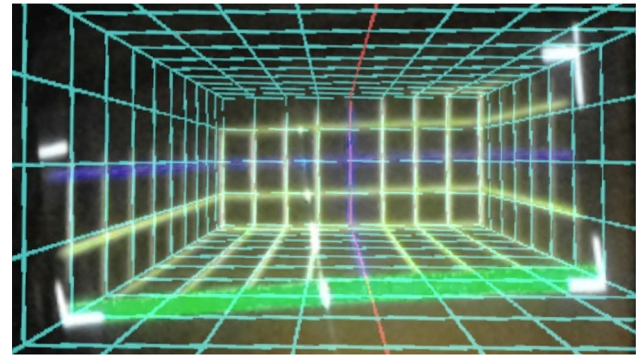
step is generating meshes for display layers. To avoid evaluating the function $\{\chi_n\}$ which is required by Eq. (6), we generate meshes $\mathcal{M}_n^{x_v}$ for each display layer that match the shape of the display layer seen at specified viewpoints $\{\mathbf{x}_v\}$. By doing so, the displayed light field L can be generated by rendering these meshes textured with the images $\{I_n\}$ instead. The meshes are generated according to the mapping function $\psi_n^{x_v}$ ($n = 1, 2; v = 1, 2$). In every frame, at the start of iteration, images $\{I_n\}$ are set to random values. After L is rendered using current images, an update operation is performed for every image using the update rule specified in Eq. (8). Then L is rendered again using updated images. This iteration is performed several times in a frame.

6.3 Calibration Process

The calibration of HUD contains three steps. As a preparation step, the camera intrinsic parameters are measured and a tracking system is setup to make the camera continuously trackable. We determine the intrinsic parameters by the standard chessboard-based calibration method proposed in [35]. Opti-Track system is used as the tracking system in our experiments, with markers stuck on windshield and camera. But devices with camera and inside-out tracking capability are also applicable (e.g., HoloLens). Then hundreds of frames from various viewpoints in eyebox are captured during the



(a)



(b)

Fig. 9 Two example scenes show the results of calibration and image correction

movement of camera while the HUD showing the grid pattern. A fast grid pattern detection method (Fig. 6) is designed to extract cross points $\{q_i\}$ from captured frames in real time, which are then converted into $\{u_i\}$ using Eq. (9) to form the training dataset. Finally, given the training dataset, we solve Eq. (14) using nonlinear fitting method provided by Matlab.¹ Cross-validation is used to avoid over-fitting. The results $\{b_{p_i}(\cdot)\}$ are saved to file for runtime use.

7 Results

7.1 Calibration

In this section, we describe the experiments we have done to evaluate the precision of our proposed calibration approach and show the result. To train and test the model, we take two sequences of samples: the sequence used as train data contains 963 points with range of $155.6 \text{ mm} \times 68.3 \text{ mm} \times 391.1 \text{ mm}$ and the sequence used as test data contains 841 points with range of $119.4 \text{ mm} \times 72.6 \text{ mm} \times 372.0 \text{ mm}$. The viewing positions of these two sequences are plotted in Fig. 7. The pattern displayed on the

¹ <https://www.mathworks.com/products/matlab.html>.

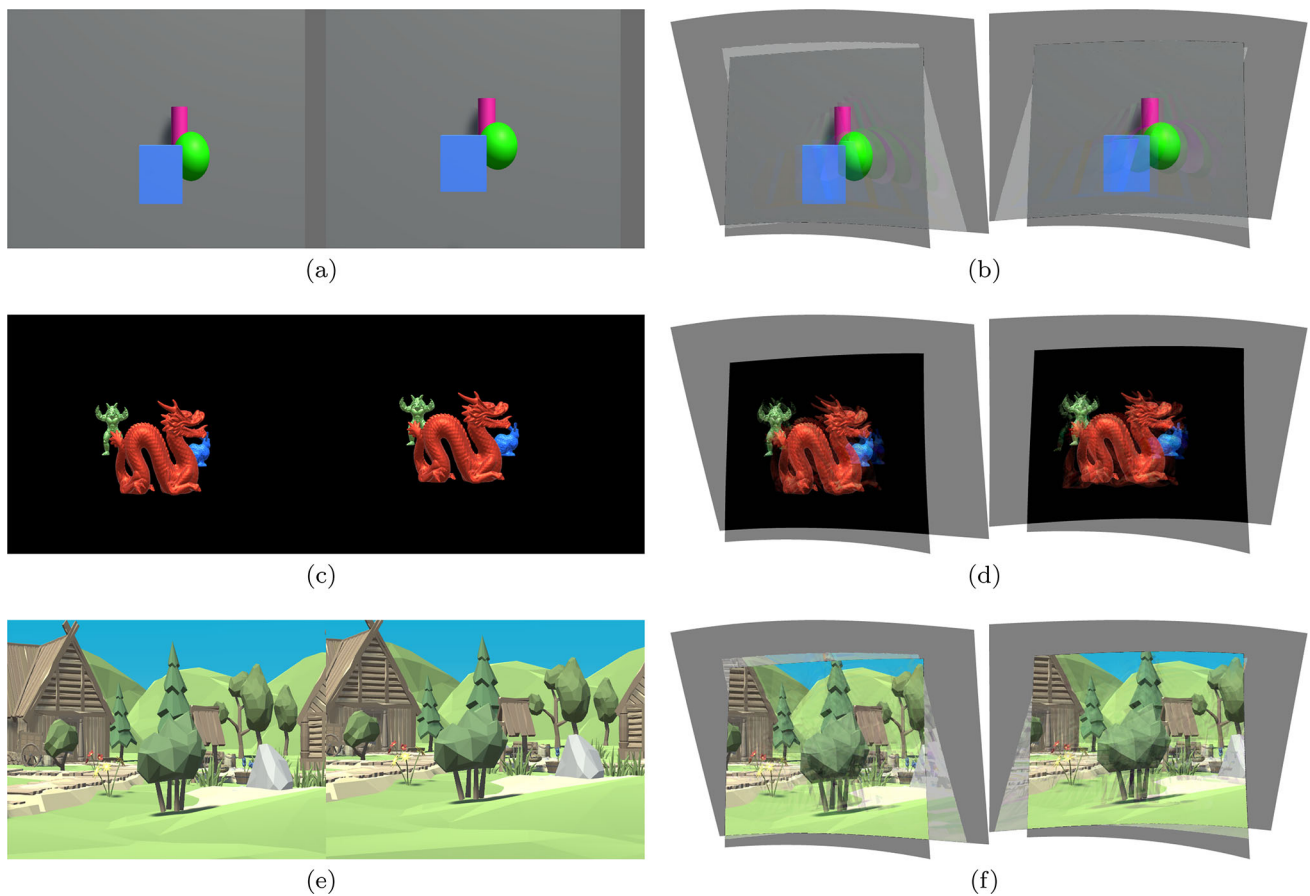
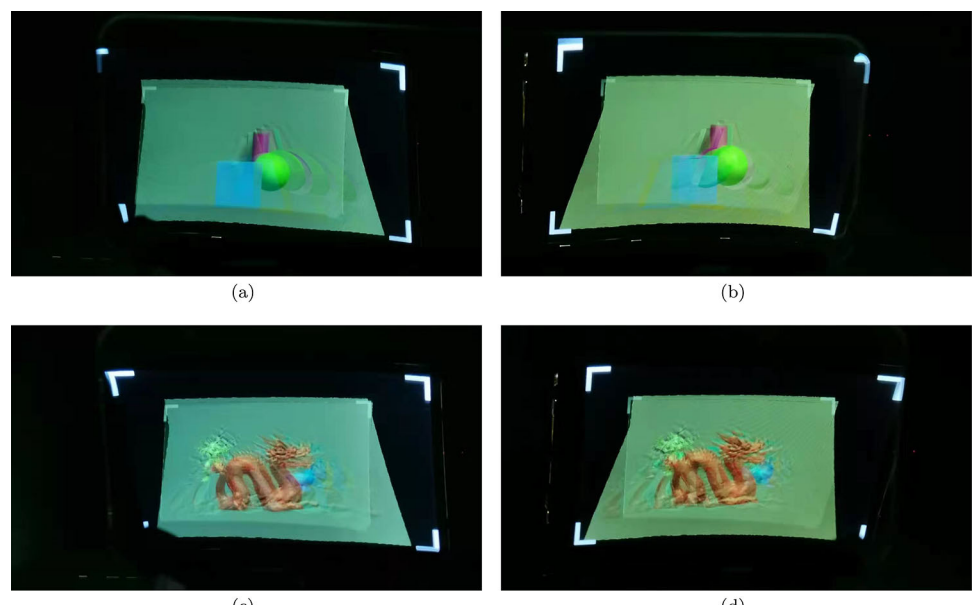


Fig. 10 Factorization results. **a, c, e** are the target light fields of three demo scenes. **b, d, f** are the corresponding reconstructed light fields

Fig. 11 Stereoscopic images of the scenes "Primitive" and "Stanford" shown on our prototype. **a** and **c** are captured at the position of left eye. **b** and **d** are captured at the position of right eye



HUD has 13 horizontal lines and 21 vertical lines, forming 273 cross points.

Polynomial Degree To compare models with different polynomial degree, we train models using train data with D taken from 1 to 5. Figure 8 shows the RMSE (root-mean-square error) of models evaluated using train data and test data. The result shows that although the error by train data keeps decreasing with higher polynomial degree, the minimum point is at $D = 2$ for error by test data, which is 6.6 mm. This indicates an over-fitting of models with $D > 2$. As a result, we infer that calibration model with $D = 2$ is most suitable for tested HUD device. What should be noticed is that the optimal D is related to the optical characteristic of HUD, so for different structure, it may be different.

Image Correction To evaluate the image correction process, we built two standard scenes: grid floor and grid tunnel. The grid floor contains horizontal and vertical lines in the $Y = 0$ plane, with distance up to 100 meters. The grid tunnel is 4 meters width and 2 m high, with a grid wall at $Z = 30$ m. Same scenes are displayed both in the AR-HUD and a monitor application simultaneously. The position of camera is used as the viewing position. Figure 9 shows the results of these two scenes at different viewpoints.

7.2 Factorization Results

Figure 10 shows the factorization results of three demo scenes. The first scene contains three primitive objects with gray background. The second scene contains three Stanford scan models (Dragon, Bunny, and Armadillo) with black background. The third scene is a poly-style village scene contains near trees and far houses. Figure 11 shows the initial results on our stereoscopic AR-HUD prototype.

8 Conclusion

This paper proposed a novel on-vehicle AR-HUD prototype that enables stereoscopic images without wearing equipment such as glasses. It is composed of two traditional HUDs and produces a two-view light field by additive light field factorization. The structure of the prototype is simple, which makes it much easier to be applied in practice. To solve the factorization problem with distortion, a render-based factorization method and an end-to-end model for HUD were presented. The proposed device naturally has less brightness problem than those general AR-HUDs because the display layers of the device are combined in an additive manner, thus doubling its brightness.

There still exist several problems to solve. First, the difference of color gamut between the HUD's display layers is significant. A color calibration for these display layers is necessary. Second, obvious artifacts like ghosts exist around

the border of virtual objects. Additional constraints should be added to suppress this effect. Third, a minor shift of the viewer's position will cause image degradation, but the viewer can hardly put eyes precisely at specific viewpoints. Future work will be expanding the sweet point region to improve the image quality in practice. Yet, the result of this paper shows a good start for implementing a practical stereoscopic HUD.

Revision Report

First of all, we greatly appreciate the reviewers for their valuable comments. According to the reviewers' comments, we have done the following revision:

- Write more clearly in Sect. 1 and 8 that the display layers of the proposed HUD are combined in an additive manner (r1);
- Add more discussion about the issues of state-of-the-art AR-HUDs in Sect. 1 (r2);
- Add a discussion about the brightness of our system in Sect. 8 (r3);
- Add citation: Xia et al. [31]. Towards efficient 3D calibration for different types of multi-view autostereoscopic 3D displays, CGI 2018. in Sect. 2 (r1);
- Carefully revise the written problem in the paper. (r1, r2, r3)

For other comments from the reviewers, we have the following answers:

- The result images captured from the left and right eyes are shown in Figs. 1 and 11 (r1)
- The need of stereoscopy as a depth cue in vehicle HUD is discussed in the second paragraph of Sect. 1 (r3)
- Existing problems which cause the artifacts in current results and also the possible solutions are discussed in Sect. 8. (r3)

Declarations

Conflict of interest Disclosure of potential conflicts of interest: No.

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